

1 Genome sequencing and assembly

Genomic DNA was extracted with the GP1 method. The harvested DNA was detected by the agarose gel electrophoresis and quantified by Qubit. The genome of XXX was sequenced by Single Molecule, Real-Time (SMRT) technology. Sequencing was performed at the Beijing Novogene Bioinformatics Technology Co., Ltd. The low quality reads were filtered by the SMRT Link v8.0 the filtered reads were assembled using software Falcon to generate contigs .

2 Genome Component prediction

Genome component prediction included the prediction of the coding gene, repetitive sequences and non-coding RNA. The available steps were proceeded as follows:

1) For Fungi, by default, we used the Augustus 2.7 program to retrieve the related coding gene. If homology reference gene sequences and transcript sequencing data were provided , a complete annotation pipeline, PASA, as implemented at the Broad Institute, involves the following steps:(A) ab initio gene finding using a selection of the following software tools: GeneMarkHMM, FGENESH, Augustus, and SNAP, GlimmerHMM.(B) protein homology detection and intron resolution using the GeneWise software and the uniref90 non-redundant protein database.(C) alignment of known ESTs, full-length cDNAs, and most recently, Trinity RNA-Seq assemblies to the genome.(D) PASA alignment assemblies based on overlapping transcript alignments from step (C). (E) use of EVidenceModeler (EVM) to compute weighted consensus gene structure annotations based on the above (A, B, C, D). (F) use of PASA to update the EVM consensus predictions, adding UTR annotations and models for alternatively spliced isoforms (leveraging D and E).

2) The interspersed repetitive sequences were predicted using the RepeatMasker (<http://www.repeatmasker.org/>). The tandem Repeats were analyzed by the TRF (Tandem repeats finder).

3) Transfer RNA (tRNA) genes were predicted by the tRNAscan-SE . Ribosome RNA (rRNA) genes were analyzed by the rRNAmmer . sRNA , snRNA and miRNA were predicted by BLAST against the Rfam database.

3 Gene function

We used seven databases to predict gene functions. They were respective GO (Gene Ontology), KEGG (Kyoto Encyclopedia of Genes and Genomes), KOG (Clusters of Orthologous Groups), NR (Non-Redundant Protein Database databases), TCDB (Transporter Classification Database), P450, and, Swiss-Prot . A whole genome Blast search (E-value less than $1e-5$, minimal alignment length percentage larger than 40%) was performed against above seven databases. The secretory proteins were predicted by the Signal P database. Meanwhile, we analyzed the secondary metabolism gene clusters by the antiSMASH . For pathogenic fungi, we added the pathogenicity and drug resistance analyses. We used the PHI (Pathogen Host Interactions), DFVF(database of fungal virulence factors) to perform the above analyses. Carbohydrate-Active enzymes were predicted by the Carbohydrate-Active enZymes Database .

4 Comparative genomics analysis

Comparative genomic analysis included the genomic synteny, the core genes and specific genes, gene family phylogenetic tree, SNP (Single Nucleotide Polymorphism), indel (insertion and deletion) and SV(Structural Variation) annotation, and genome visualization. 1) Genomic alignment between the sample genome and reference genome (or among more than two sample genomes) were performed using the MUMmer and LASTZ tools. Genomic synteny was analyzed based on the alignment results. 2) Core genes and specific genes were analyzed by the CD-HIT rapid clustering of similar proteins software with a threshold of 50% pairwise identity and 0.7 length difference cutoff in amino acid. Then the venn figure was drawn to show their relationships among the samples. 3) Gene family was constructed by the gene of XXX and XXX, using the multiple softwares: Blast was used to pairwise align all genes and eliminate the redundancy by solar and carried out gene family clustering treatment based on the alignment results with Hcluster_sg software. 4) The phylogenetic tree was constructed by the TreeBeST or PhyML and the setting of bootstraps was 1,000 with the orthologous genes. 5) SNP, indel and SV were found by the genomic alignment results among samples by the MUMmer and LASTZ we mentioned them before.